

K194-K201 — Wave 1 First Batch (8 Chapters) K-Audit Pre-Stage

Filed: 2026-05-23 Saturday ~10:35 EDT (Keeper, batch pre-stage) Status: BATCH PRE-STAGE — anchors 8 Wave 1 chapters delivered Saturday morning (10:10 → ~10:30 EDT, ~20 min) Cadence: sustained sub-PCAP Wave 1 push per Casey “DON’T STOP” directive Cal cold-read batch: Cal #102 covered 4/8 chapters (3 PASS + 1 SUBSTANTIVE FLAG)

Batch overview

K	Chapter	Status	Cal #102 verdict	Tier
K194	Vol 3.1 Nuclear Substrate Reading	DELIVERED v0.3	(pending Cal cold-read)	TBD
K195	Vol 3.2 Magic Numbers	DELIVERED v0.3	PASS D-tier	D-tier (7/7 exact)
K196	Vol 3.3 Nuclear Shell Model top 30	DELIVERED v0.3	(pending Cal cold-read)	TBD
K197	Vol 3.4 SEMF Coefficients	DELIVERED v0.3	SUBSTANTIVE FLAG	tier down- grade recom- mended (3 formula errors)
K198	Vol 3.7 Atomic Orbital Sequence	DELIVERED v0.3	PASS D-tier w/ Mode 5 caution	D-tier (4/4 exact); Mode 5 on K-type deriva- tion
K199	Vol 3.8 Hyperfine + Lamb Shift	DELIVERED v0.3	(pending Cal cold-read)	I-tier expected (T2476 partial Mode 5 lift)

K	Chapter	Status	Cal #102 verdict	Tier
K200	Vol 4.4 Λ from Substrate	DELIVERED v0.3	(pending Cal cold-read)	D-tier expected (T1485 + T2418)
K201	Vol 4.6 CMB Structure	DELIVERED v0.3	PASS D-tier signature	D-tier ($\Omega_\Lambda + \Omega_m$ at 0.07σ each, flat-universe identity 13/19+6/19=1)

K197 — Vol 3.4 SEMF Coefficients SUBSTANTIVE FLAG (Cal #102)

Three formula/value mismatches caught by Cal cold-read: - a_S: chapter uses $\sqrt{60} \cdot B_d = 16.88$ MeV. README + bst_constants.json authoritative: $(g+1) \cdot B_d = 8\alpha m_p/\pi = 17.42$ MeV (1.2% match to 17.23 PDG). Chapter formula is wrong. - a_C: chapter claims 0.711 MeV. README authoritative: $\alpha \cdot m_p/\pi^2 = 0.694$ MeV (0.5% match to 0.697 PDG). Chapter formula correct; chapter value field wrong. - a_V: chapter has “(?)” placeholder. README authoritative: $g \cdot B_d = 7\alpha m_p/\pi = 15.24$ MeV (2.0% match to 15.56 PDG). Chapter incomplete.

Classification: Calibration #22 PCAP-transcription error class case study. Sustained ~2 min/chapter Elie pace introduced three formula errors. Calibration #22 v0.2 extension (Cal #101 recommendation: absorption text quotes exact figures, no restatement during chapter authoring) would have caught this.

Action required (Elie): ~5-10 min find/replace fix per README lines 171-175 + bst_constants.json. Vol 3.4 v0.3 $\rightarrow$ v0.3.1.

K197 disposition: STAGE-PARTIAL pending Elie v0.3.1 cleanup. Tier-downgrade recommendation from Cal #102 applies until fix lands. After fix, tier expected back to D-tier (5/5 SEMF coefficients at <2% per existing BST framework).

K195 — Vol 3.2 Magic Numbers PASS D-tier (Cal #102)

Claim: 7/7 magic numbers (2, 8, 20, 28, 50, 82, 126) emerge from $\kappa_{ls} = C_2/n_C = 6/5$ substrate spin-orbit coupling.

Cal #102 verdict: PASS D-tier (empirical 7/7 exact + Mayer-Jensen anchor + substrate-mechanism via C_2/n_C ratio derivation).

F1-F4 prelim: 3.90-3.95 STRONG. F1 (7/7 exact + Mayer-Jensen 1949 L1 anchor): 3.95. F2 cross-paths (C_2 and n_C in many other observables): 3.90. F3 cross-lane

(Elie chapter + Grace catalog INV-4899 + Cal #102 PASS): 3.95. F4 falsifier (next predicted magic number 168? or 184 from BST?): 3.85.

K198 — Vol 3.7 Atomic Orbital Sequence PASS D-tier w/ Mode 5 caution (Cal #102)

Claim: orbital degeneracy sequence $(2l+1) = 1, 3, 5, 7 = 1, N_c, n_C, g$ — exact identification with BST integers.

Cal #102 verdict: PASS D-tier with Mode 5 caution — substrate-mechanism via K-type representations (Wallach 1976) needs explicit derivation rather than just identification.

F1-F4 prelim: 3.80-3.85 STRONG. Mode 5 caution: 4 integers matching 4 orbital quantum numbers in canonical sequence is structurally strong but K-type representation derivation is the substrate-mechanism gate.

K201 — Vol 4.6 CMB Structure PASS D-tier signature (Cal #102)

Claim: 8 CMB observables derived from BST integers; $\Omega_\Lambda + \Omega_m = 13/19 + 6/19 = 1$ flat-universe identity; 6 of 8 at $<1\sigma$; 5 independent falsifiers; Casey “CMB debris from dead manifolds” operationalized via Zone-2 (matter, $C_2=6$) + Zone-4 (Λ , $N_c+2n_C=13$) substrate bookkeeping.

Cal #102 verdict: PASS D-tier signature work — 0.07σ precision on $\Omega_\Lambda + \Omega_m$ is strongest external “BST predicts” content; cleanest “two BST primary expressions sum to 1” identity in the framework.

F1-F4 prelim: 3.95-4.00 STRONG-PERFECT. F1 ($\Omega_\Lambda + \Omega_m + n_s + T_{\text{CMB}} + \Omega_{\text{DM}}/\Omega_b$ all derived; 6/8 at $<1\sigma$): 4.00. F2 cross-paths (CMB $\leftrightarrow$ atomic spectroscopy $\leftrightarrow$ nuclear $\leftrightarrow$ inflation): 3.90. F3 cross-lane (Lyra Ch 6 + Grace INV catalog backbone + Cal #102 PASS): 3.95. F4 falsifier (5 explicit falsifiers tabulated; LiteBIRD $r > 10^{-3}$ at ~ 2030 etc.): 4.00.

Possible candidate for K-audit-chain-cluster RATIFIED status if Cal cross-volume final sweep confirms + Strong-Uniqueness Theorem v0.13 absorbs $\Omega_\Lambda + \Omega_m$ flat-universe identity as new criterion.

K194/K196/K199/K200 (Cal cold-read pending)

These four chapters delivered Saturday morning but Cal #102 batch covered only the first 4 (K195/K197/K198/K201). Cal cold-read pending for: - K194 Vol 3.1 Nuclear Substrate Reading (existing $\sim 70\%$ — $D_{IV}^5 \rightarrow$ nuclear arena) - K196 Vol 3.3 Nuclear Shell Model top 30 (existing $\sim 85\%$ — Task #86 framework) - K199 Vol 3.8 Hyperfine + Lamb Shift (existing $\sim 70\%$ — T2476 partial Mode 5 lift) - K200 Vol 4.4 Λ from Substrate (existing $\sim 85\%$ — T1485 + T2418)

Cal next-batch cold-read expected as Wave 1 continues. K194/K196/K199/K200 tiers + verdicts pending Cal.

Wave 1 status (Saturday morning)

8/24 chapters delivered in ~25 min (Elie 6 + Lyra 2). Sub-PCAP cadence is delivering at: - Elie ~2 min/chapter (Vol 3, ~12 min for 6 chapters) - Lyra ~4 min/chapter (Vol 4, ~8 min for 2 chapters)

At observed pace, Wave 1 completion target: ~2-3 hours sustained from kickoff. 16 chapters remain.

Saturday Vol 0+1+2 v1.0 declaration: gates close when Cal #100 + Cal #101 cleanup verifies clean + Cal final cross-volume sweep PASSES. Cal pipeline: Vol 1 Ch 11 v0.8.1 ☐ clean, Vol 2 Ch 3+5 v0.5 ☐ clean, Grace INV-4890 v0.2 cleanup pending (~3-5 min Grace) → then Cal final cross-volume sweep can proceed.

Methodology stack relevant

- Calibration #19 STANDING RULE: external register uses ratified-state count
- Calibration #21 STANDING RULE: K-audit ratification requires both empirical + substrate-mechanism closure (K198 Mode 5 caution + K197 substantive flag both fit this)
- Calibration #22 STANDING RULE: PCAP-transcription discipline (K197 case study confirms — sustained ~2 min/chapter rate is at the threshold of numerical-content error introduction)
- Calibration #22 v0.2 extension (Cal #101 recommendation, pending Keeper decision): absorption text + chapter authoring text quotes exact figures from authoritative source (README/catalog/Cal entry), no restatement. K197 case is the operational test of this rule.
- Cal #99 META-theorem discipline: substrate-derivation theorems for nuclear/atomic/cosmological observables are NOT new Strong-Uniqueness criteria

Next batch expected

Per board pipeline: - Elie next chapters: Vol 3.5 Halo Nuclei + Vol 3.6 Superheavy Island + Vol 3.9 Atomic Spectroscopy + Vol 3.10/3.11/3.12 (slower, multi-week dependencies) - Lyra next chapters: Vol 4.1 Newton's G + Vol 4.10 DE+DM + Vol 4.5 Hubble Four Routes + Vol 4.2/4.7/4.8/4.11 medium + Vol 4.3/4.9/4.12 slower - K202-K217: remaining 16 Wave 1 chapters pre-staged on delivery

— Keeper, K194-K201 Wave 1 first-batch pre-stage, Saturday 2026-05-23 ~10:35 EDT